Python Control Flow - Internship Report

1. Objective

The objective is to learn conditional statements and loops by writing Python programs.

2. Conditional Statements

The if statement executes when a condition is true. The elif statement checks another condition, and else handles the remaining case. The modulo operator (%) helps identify even and odd numbers.

3. For Loop and While Loop

A for loop iterates over a sequence or range. This project uses it for multiplication tables and triangle patterns. A while loop repeats while its condition is true. It calculates the sum of natural numbers by adding values from 1 through N.

4. Mathematical Operations

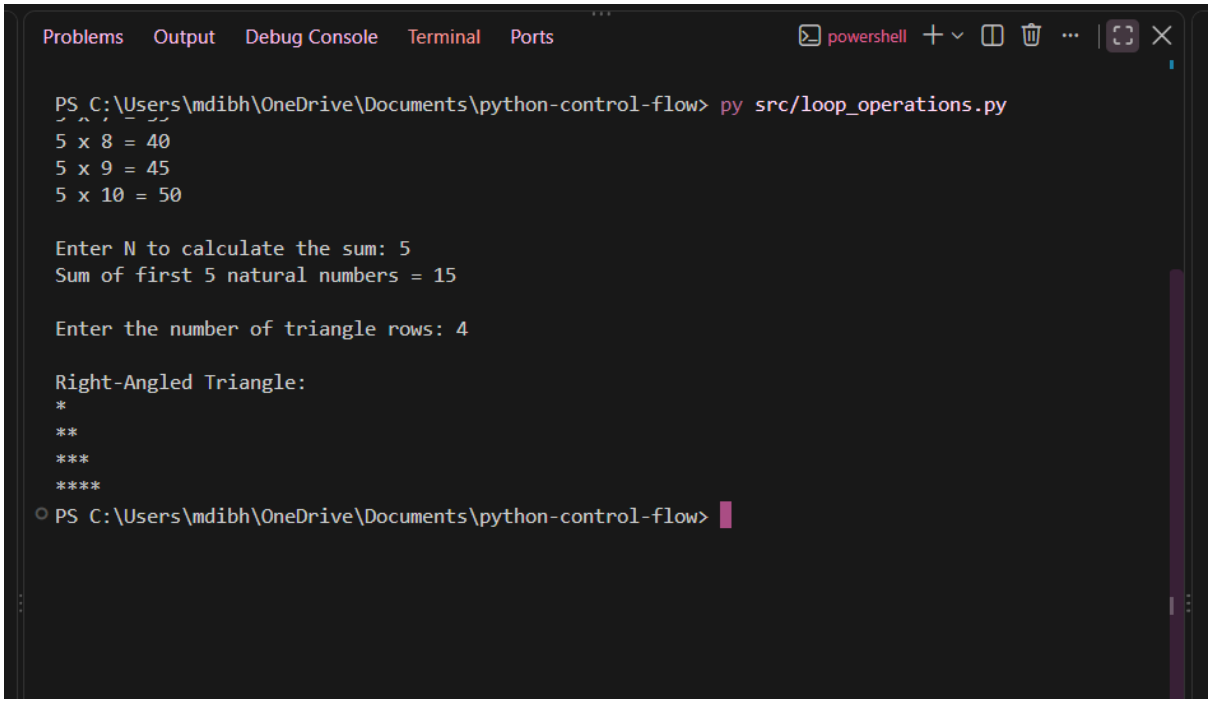
The multiplication table multiplies a selected number by integers from 1 to 10. The sum of the first five natural numbers is 1 + 2 + 3 + 4 + 5 = 15. The triangle pattern adds one asterisk on each successive row.

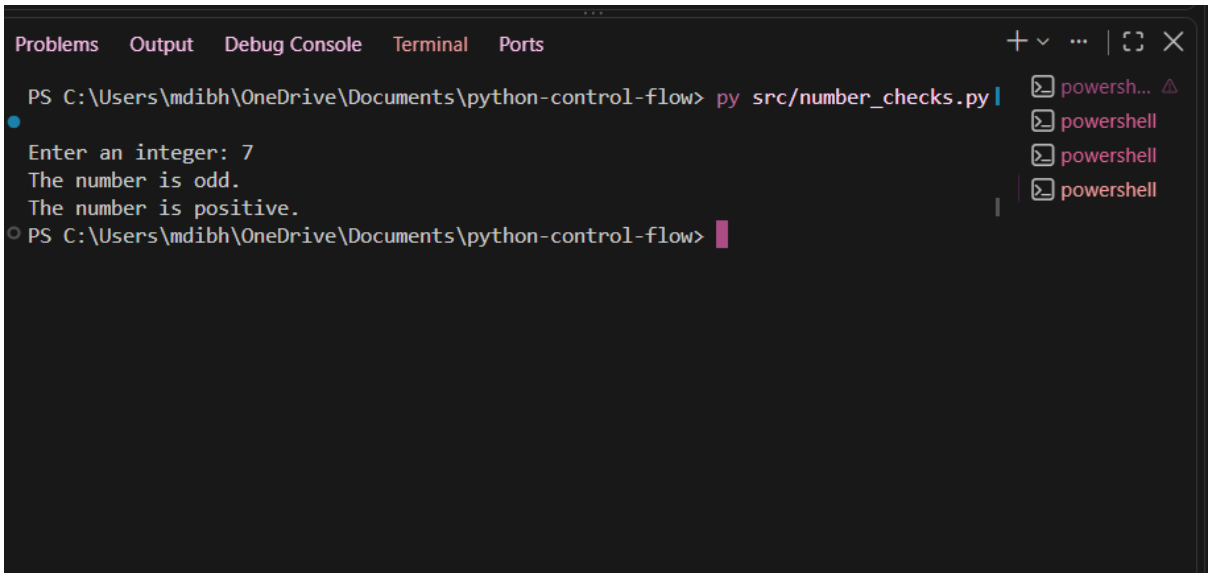
5. Testing

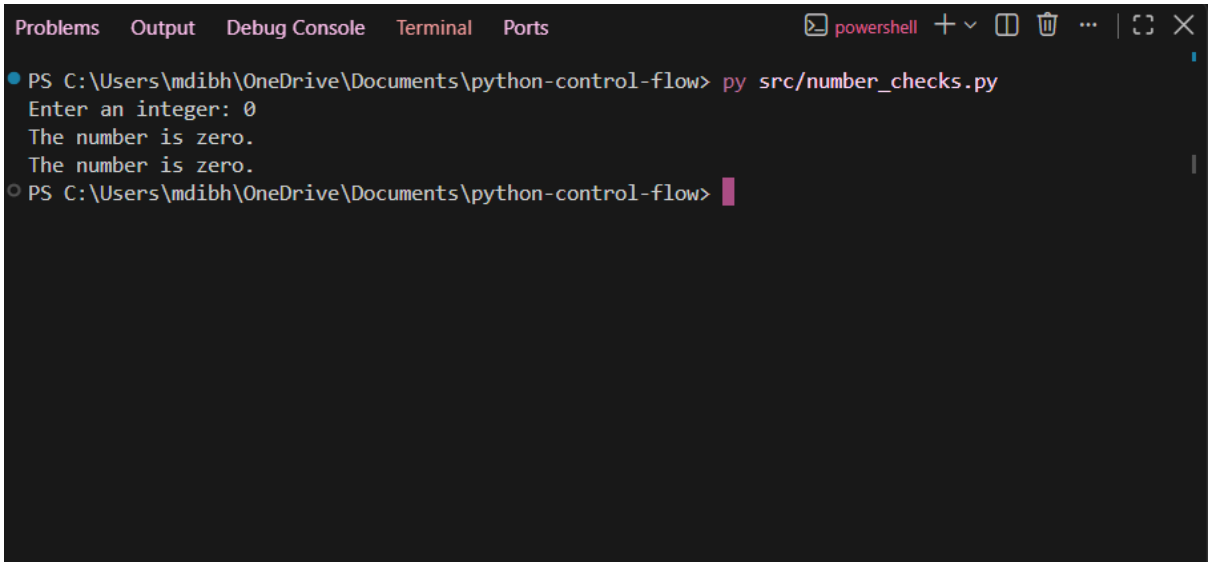
Test the number checker with 8, -5, -4 and 0. Test the loop program with different numbers, positive N values and different row counts. Non-positive N and row values are handled with messages.

6. Conclusion

This task demonstrates decision-making, loops, arithmetic, input and output in Python. Actual terminal screenshots should be added to docs/screenshots.


Problems Output Debug Console Terminal Ports
powershell
PS C:\Users\mdibh\OneDrive\Documents\python-control-flow> py src/loop_operations.py
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
Enter N to calculate the sum: 5
Sum of first 5 natural numbers = 15
Enter the number of triangle rows: 4
Right-Angled Triangle:
*
**
***
****
PS C:\Users\mdibh\OneDrive\Documents\python-control-flow>


Problems Output Debug Console Terminal Ports
PS C:\Users\mdibh\OneDrive\Documents\python-control-flow> py src/number_checks.py
Enter an integer: 7
The number is odd.
The number is positive.
PS C:\Users\mdibh\OneDrive\Documents\python-control-flow>
powersh...
powershell
powershell
powershell


Problems Output Debug Console Terminal Ports
powershell
PS C:\Users\mdibh\OneDrive\Documents\python-control-flow> py src/number_checks.py
Enter an integer: 0
The number is zero.
The number is zero.
PS C:\Users\mdibh\OneDrive\Documents\python-control-flow>